



Cold-Formed Steel Specialty Conference

Recent updates to ANSI MH16.1: perforated rack member design and the Direct Strength Method

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Three editions of MH16.1 in eleven years

- **ANSI MH16.1** (RMI) is the specification for industrial steel storage racks, referenced by ASCE 7 §15.5.3 and the IBC
- Rack uprights are roll-formed, thin, open, and **perforated along their full length**, so local, distortional, and global buckling interact
- The 2021 edition rewrote the member design chapter around elastic buckling analysis; the 2023 edition kept it
- This talk: what changed from 2012 to 2023 and where the Direct Strength Method enters



Effective area and the stub-column Q factor

Compression (§4.1.3.1)

- Effective area from the stub-column test:

$$A_e = \left[1 - (1 - Q) \left(\frac{F_n}{F_y} \right)^Q \right] A_{net\ min}$$

- $P_n = A_e F_n$, with F_n from AISI S100 on the **gross, unperforated** section
- $Q = \text{stub column strength} / (F_y A_{net\ min}) \leq 1$ (§9.2.2)

Flexure (§4.1.2)

- $S_e = S_{net\ min} (0.5 + Q/2)$
- Round corners for A and I ; sharp corners permitted for J , C_w , r_o

Distortional (§4.1.3.2)

- One sentence: certain open sections "shall be checked ... by testing or rational analysis"
- No equations, no elastic buckling load

Frame stability by the effective length method: down-aisle $K_x = 1.7$ for unbraced racks, $K_t = 0.8$

Perforations become reduced-thickness strips

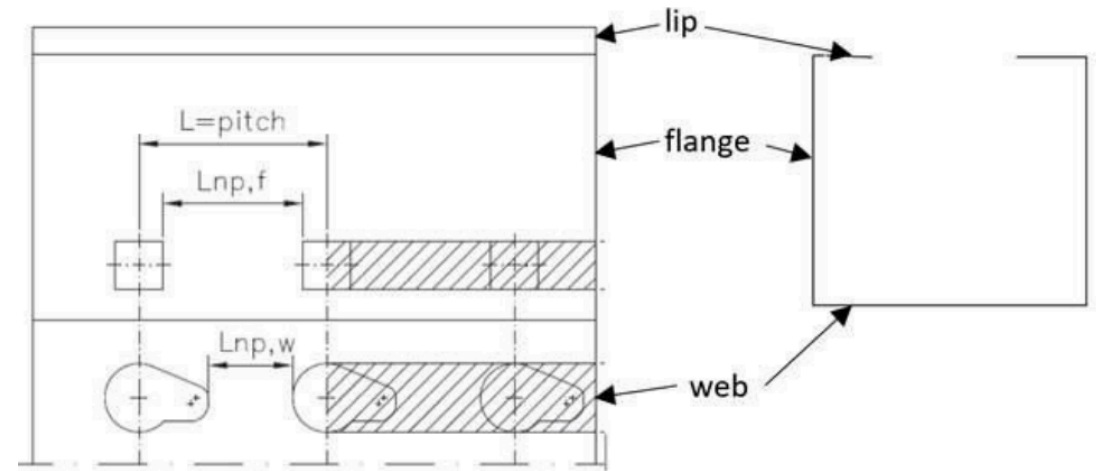
- Section properties use **round corners** for everything, including J and C_w (sharp corners are unconservative for torsion)
- Each strip of web or flange that contains perforations is replaced by a solid strip of reduced thickness
- Global and local** buckling, $k_g = 0.6$:

$$t_g = k_g t \left(\frac{L_{np}}{L} \right)$$

- Distortional** buckling, $k_d = 0.8$:

$$t_d = k_d t \left(\frac{L_{np}}{L} \right)^{1/3}$$

- L_{np} is the solid length between holes, L the pitch



Source: ANSI MH16.1-2023, Figure 8.2-1, example of perforation strips in a column

- One idealized section feeds the frame model (§7.2.4), the global buckling check, and the finite strip model for P_{crd} and M_{crd}

Perforated compression members

Global (8.2-3, 8.2-4)

$$P_{ne} = \begin{cases} 0.658^{\lambda_c^2} P_y & \lambda_c \leq 1.5 \\ \frac{0.877}{\lambda_c^2} P_y & \lambda_c > 1.5 \end{cases} \quad \lambda_c = \sqrt{P_y / P_{cre}}$$

$P_y = F_y A_{netg}$; P_{cre} with K , L from §10.2, §10.3

Local + global (8.2-5, 8.2-6)

$$P_{nlg} = P_{ne} \left[1 - (1 - Q) \left(\frac{P_{ne}}{P_y} \right)^{\frac{Q}{1-Q}} \right], \quad Q < 1$$

$P_{nlg} = P_{ne}$ if $Q \geq 1$ (split out as its own equation in 2023)

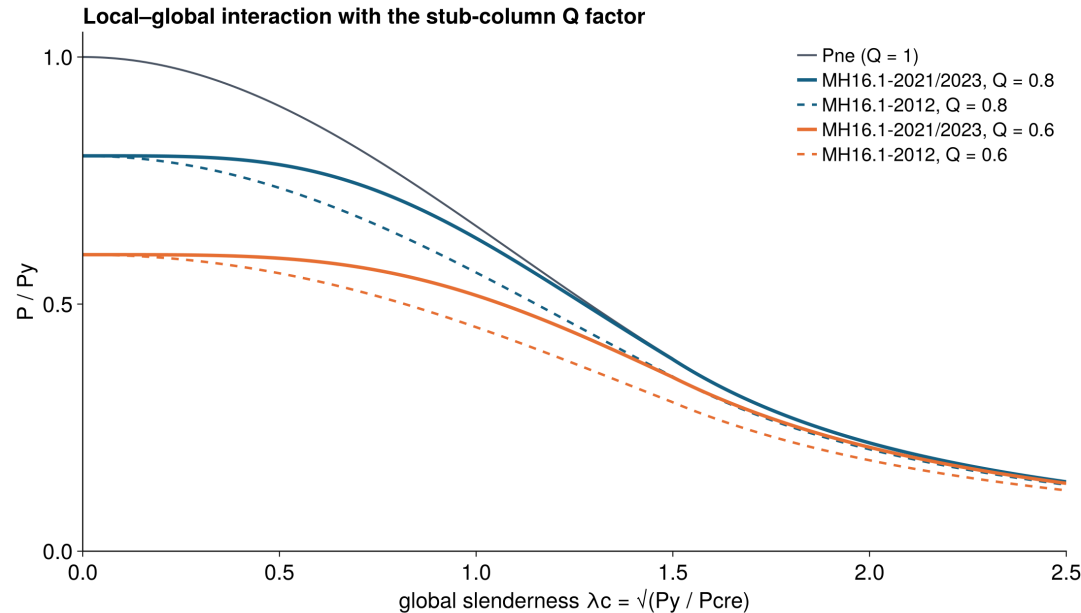
Distortional + global (8.2-7, 8.2-8)

$$P_{nld} = \left[1 - 0.25 \left(\frac{P_{crd}}{P_{ne}} \right)^{0.6} \right] \left(\frac{P_{crd}}{P_{ne}} \right)^{0.6} P_{ne}$$

for $\lambda_d = \sqrt{P_{ne} / P_{crd}} > 0.561$, else $P_{nld} = P_{ne}$

- $P_n = \min(P_{nlg}, P_{nld})$
- P_{crd} : elastic distortional buckling "in accordance with ANSI/AISI S100" on the t_d section
- Hot-rolled and closed sections: P_{nlg} only

Still the stub column, in a new form

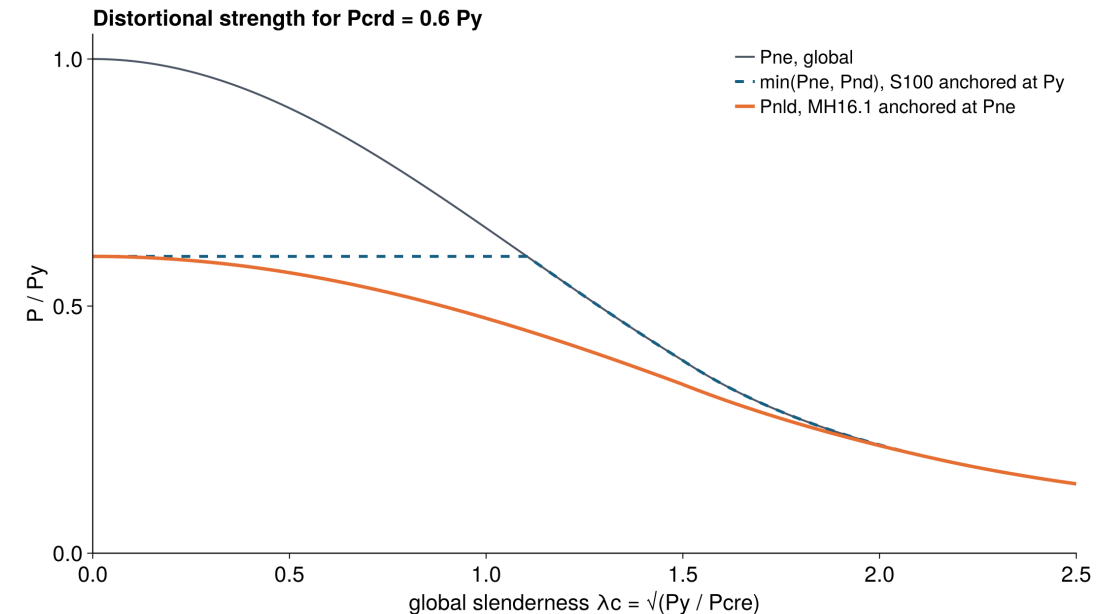


Both forms with $F_n/F_y = P_{ne}/P_y$ on the same area basis. 2012 used $A_{net\ min}$; 2021 uses A_{netg} from the strip section.

- Local buckling is **not** a DSM check: there is no $P_{cr\ell}$. The perforation and local slenderness effects both come from the **stub-column test** (§13.2.3, $Q = P_{test}/(F_y A_{netg}) \leq 1$, interpolated between t_{min} and t_{max})
- The 2021 exponent $Q/(1 - Q)$ makes the local penalty fade as the column becomes globally slender, the same trend as the DSM local-global curve
- For a given Q , the new form is less severe at intermediate λ_c

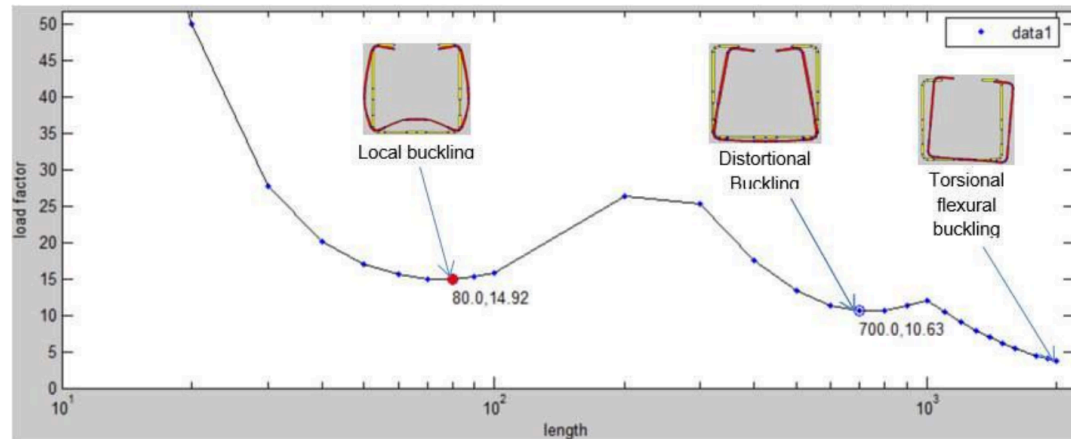
The Direct Strength Method piece

- New in 2021: an explicit distortional check with an elastic buckling load, P_{crd} or M_{crd} , the DSM ingredient
- Same 0.25 / 0.6 coefficients and 0.561 limit as the AISI S100 distortional curve, but **anchored at P_{ne} instead of P_y**
- That captures distortional–global interaction, which tests and FE studies on perforated uprights showed (Casafont, Pastor, Roure, Bonada, Peköz)
- At $\lambda_c = 0$ the two agree; at intermediate slenderness MH16.1 is lower, and both reach P_{ne} for slender columns



Example with $P_{crd} = 0.6 P_y$

Finite strip analysis on the strip section



Source: ANSI MH16.1-2023 Commentary, Figure C8.2-5. CUFSM: local minimum at 80 mm (load factor 14.92), distortional at 700 mm (10.63)

- Commentary C8: the "most suitable design approach would be to use finite strip analysis combined with the expressions given"
- Workflow
 - build the round-corner section with t_d in the perforated strips
 - run CUFSM (or any S100 Appendix 2 analysis)
 - read P_{crd} at the distortional minimum
 - the same model with t_g gives the global section properties
- Basis: tests, shell FE and FSM studies on perforated uprights (Casafont et al., ASCE *J. Struct. Eng.* 2013, "Design of steel storage rack columns via the Direct Strength Method")

Flexure follows the same pattern

Local + global (8.2-9, 8.2-10)

$$M_{nlg} = M_{ne} \left[1 - \frac{1}{2}(1 - Q) \left(\frac{M_{ne}}{M_y} \right)^{\frac{Q}{1-Q}} \right], \quad Q < 1$$

$M_y = F_y S_{fy}$, with S_{fy} on the reduced-thickness section

Distortional + global (8.2-11, 8.2-12)

$$M_{nld} = \left[1 - 0.22 \left(\frac{M_{crd}}{M_{ne}} \right)^{0.5} \right] \left(\frac{M_{crd}}{M_{ne}} \right)^{0.5} M_{ne}$$

for $\lambda_d = \sqrt{M_{ne}/M_{crd}} > 0.673$, else $M_{nld} = M_{ne}$

- $M_n = \min(M_{nlg}, M_{nld})$ for open cold-formed sections bending about the axis of symmetry; hot-rolled and closed sections use M_{nlg}
- The coefficients match the AISI S100 flexural distortional curve, anchored at M_{ne} like compression
- M_{crd} comes from finite strip analysis on the t_d section, as for P_{crd}

Frame stability: from effective length to direct analysis

- **§7.2: second-order analysis with notional loads**
replaces the effective length method
 - $N_i = 0.004 \alpha Y_i$ with the stiffness reduction τ_b , or $0.005 \alpha Y_i$ without; minimum $0.002 \alpha Y_i$
 - $B_2 = 1 / (1 - \alpha P_\Delta / (R_m H L))$
 - semi-rigid beam-to-column connections modeled with springs
- Down-aisle $K_x = 1.0$ (§10.2.1)
- Member and frame models use the same hole-reduced section properties
- **Connectors from cyclic tests** (§13.5): design moment $0.75 M_{max}$; stiffness is the secant at $0.8 M_{conn,d}$
- New tests: base fixity (§13.6) and frame bracing (§13.7); portal and upright frame tests removed
- Seismic aligned to ASCE 7-16; redundancy factor for multiple rows; overstrength design of base plates and anchors (§11.3)
- ISO-style reorganization of the whole document



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